

Satellite-Derived Shorelines

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Highlights

- There is little performance variation between RGB+NIR and RGB+NIR+SWIR approaches
- RMSE varies between sites with lower tidal range sites performing better

Motivation

Beaches and sandy coastlines are economically, culturally, and ecologically important, monitoring them is crucial. To understand coastal change, we rely on historical shoreline monitoring data. However, due to the substantial time and resources needed for consistent surveys, only a few locations worldwide have been consistently monitored over the past decades. Satellite-Derived Shorelines (SDS) provide a scalable approach to beach monitoring.



Figure 1: North Shore, O'ahu Coastal Erosion

Satellites

In this work six satellite missions are used: Landsat (5, 7, 8, 9), Sentinel-2, PlanetScope. Landsat imagery covers from 1984 to present day with monthly to bimonthly revisit times and 30 m (Landsat 5) to 15 m (Landsat 7-9) spatial resolution. Sentinel-2 has recorded imagery since June 2015 with 5-day revisit time. The PlanetScope Constellation has recorded the entire land surface of the earth with near-daily revisit time since 2014.

Top of atmosphere radiance values for red, green, blue, near-infrared, and short-wave infrared bands were downloaded. Landsat and Sentinel-2 are publicly available and can be downloaded using the following python package:

<https://github.com/Coastal-Research-Collaborative/geedownload>

PlanetScope imagery is proprietary with free educational licenses available. PlanetScope imagery can be downloaded using the following package (Note PlanetScope does not collect shortwave-infrared data):

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Figure 2: Satellite Sources Makapu'u, O'ahu

Site Description

It is important to validate SDS algorithms against in-situ beach survey data. This study evaluates five approaches against in-situ shoreline positions from six global sites. Narrabeen, Australia, is an over 3-kilometer (km) long embayed beach located on the south-eastern coast of Australia. Monthly beach profile surveys along five cross-shore transects have been conducted since 1976. Waikiki, Hawai'i is a 520 m reef-fringed, embayed beach located on the south shore of the island of O'ahu. Biweekly surveys were conducted between 2018 and 2021. The Scripps Institute of Oceanography has collected a monthly dataset at three sites in San Diego, California, between 2001-2017. Truc Vert, France is a high-energy meso-macrotidal rip-channelled beach. Monthly to fortnightly topographic surveys were conducted from September 2003 until December 2019, with a 1-year gap in 2008.

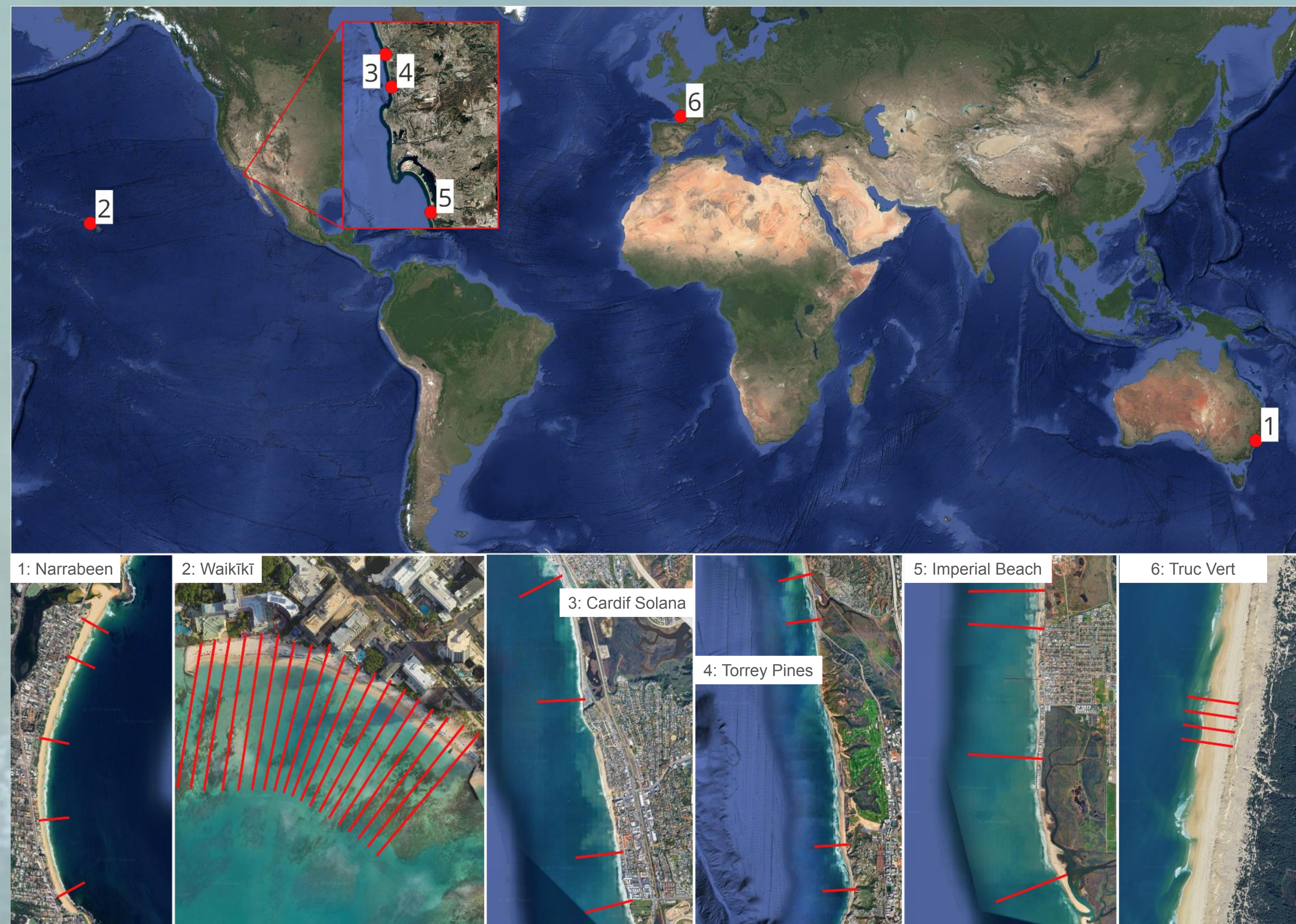


Figure 3: In-situ Site Locations and Transect Positions

Shoreline Extraction

SDS frameworks rely on segmenting the satellite image into land and water and then an algorithm such as the marching squares algorithm is used to draw the shoreline. Traditionally these approaches have mainly been index based, first a gray scale image is created using band math; such as the Normalized Difference Water Index (NDWI) (McFeeters, 1996). Once the gray scale index image is created a algorithm such as Otsu (Otsu, 1979) is used to define the threshold that best separates land and water. Recently machine learning based image segmentation has become increasingly popular. This includes pixel classifiers such as CoastSat as well as image to image models such as U-Net CNNs.

In this work three indices and two machine learning algorithms were evaluated. The indices were NDWI, Modified Normalized Difference Water Index (MNDWI), and Subtractive Coastal Water Index (SCoWI) using Otsu thresholding. The ML models were CoastSat's pixel classifier and a U-net trained on 215 hand annotated satellite images.

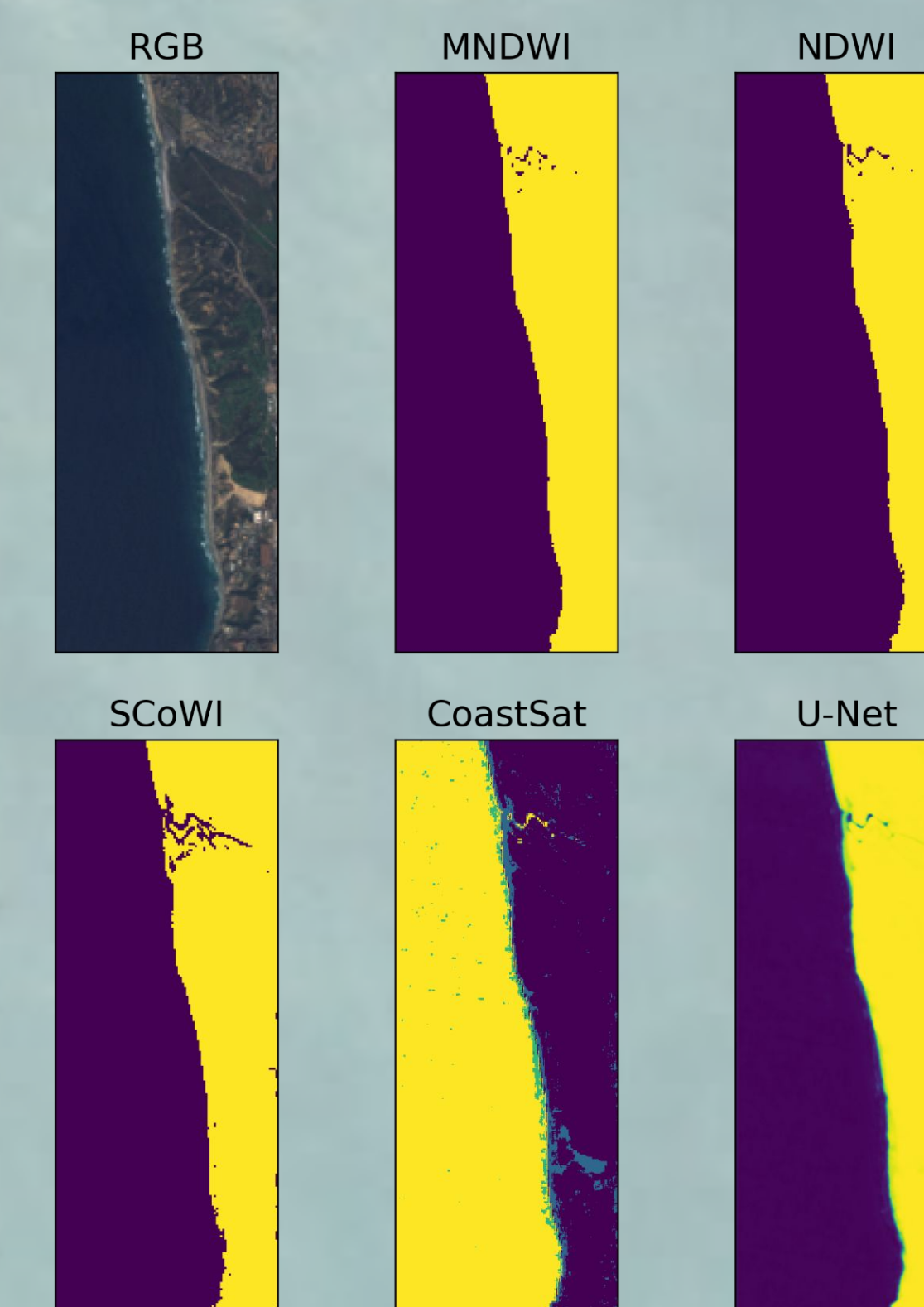


Figure 4: Example Segmentations, Torrey Pines, California

Results

Shoreline positions are traditionally measured along shore normal transects. To compare SDSs and in-situ surveys transect intersections of satellite shorelines within ten days of an in-situ are compared to in-situ intersections. It's important to note that in the current approach there is no correction for how differing tide-level and wave conditions which has been shown to improve performance (Vos et. al, 2023). Figure 5 illustrates this process on a Landsat 5 image of Narrabeen, Australia.

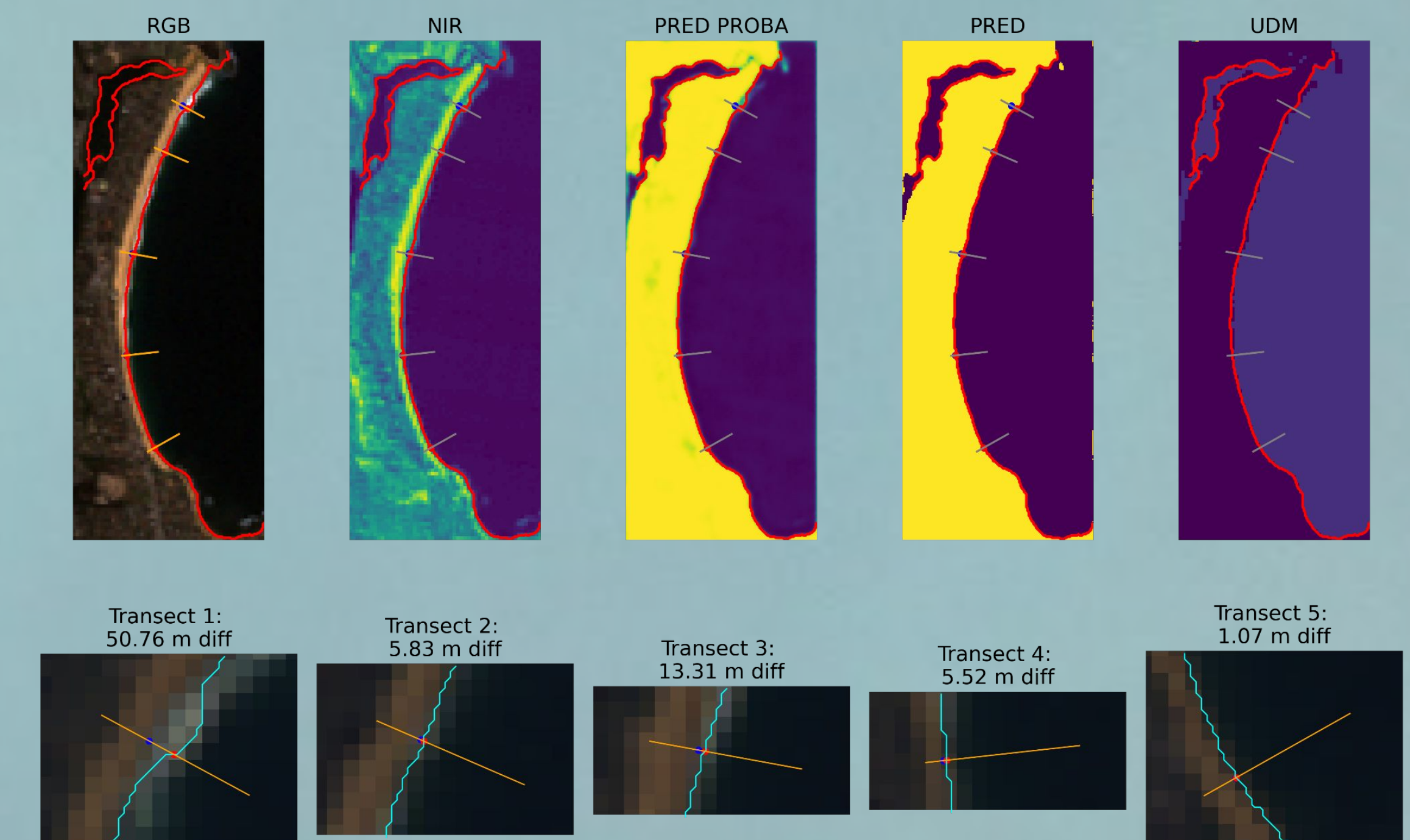


Figure 5: Transect Intersection Comparison Example, Narrabeen, Australia

- Little variation in RMSE between different segmentation approaches
- Inter-site performance is variable with Truc Vert a Macrotidal beach performing the worst



Figure 6: RMSE Against In-situ Sites, Averaged Across Satellite Mission (Excluding PlanetScope) and Transects

Acknowledgments

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References

- Vos, K. et al. Benchmarking satellite-derived shoreline mapping algorithms. Communications Earth & Environment 4 (2023). URL <https://doi.org/10.1038/S43247-023-01001-2>.
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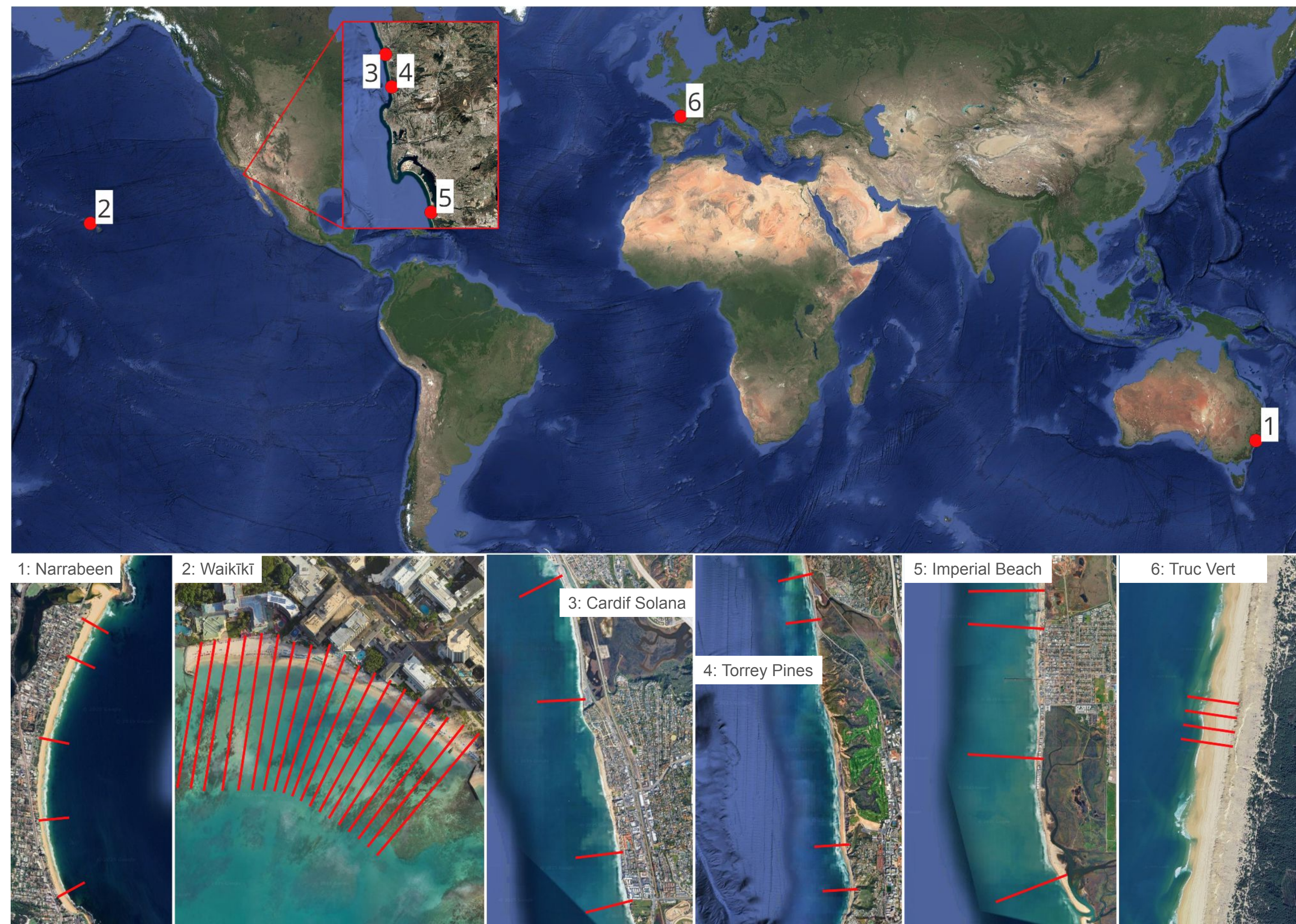


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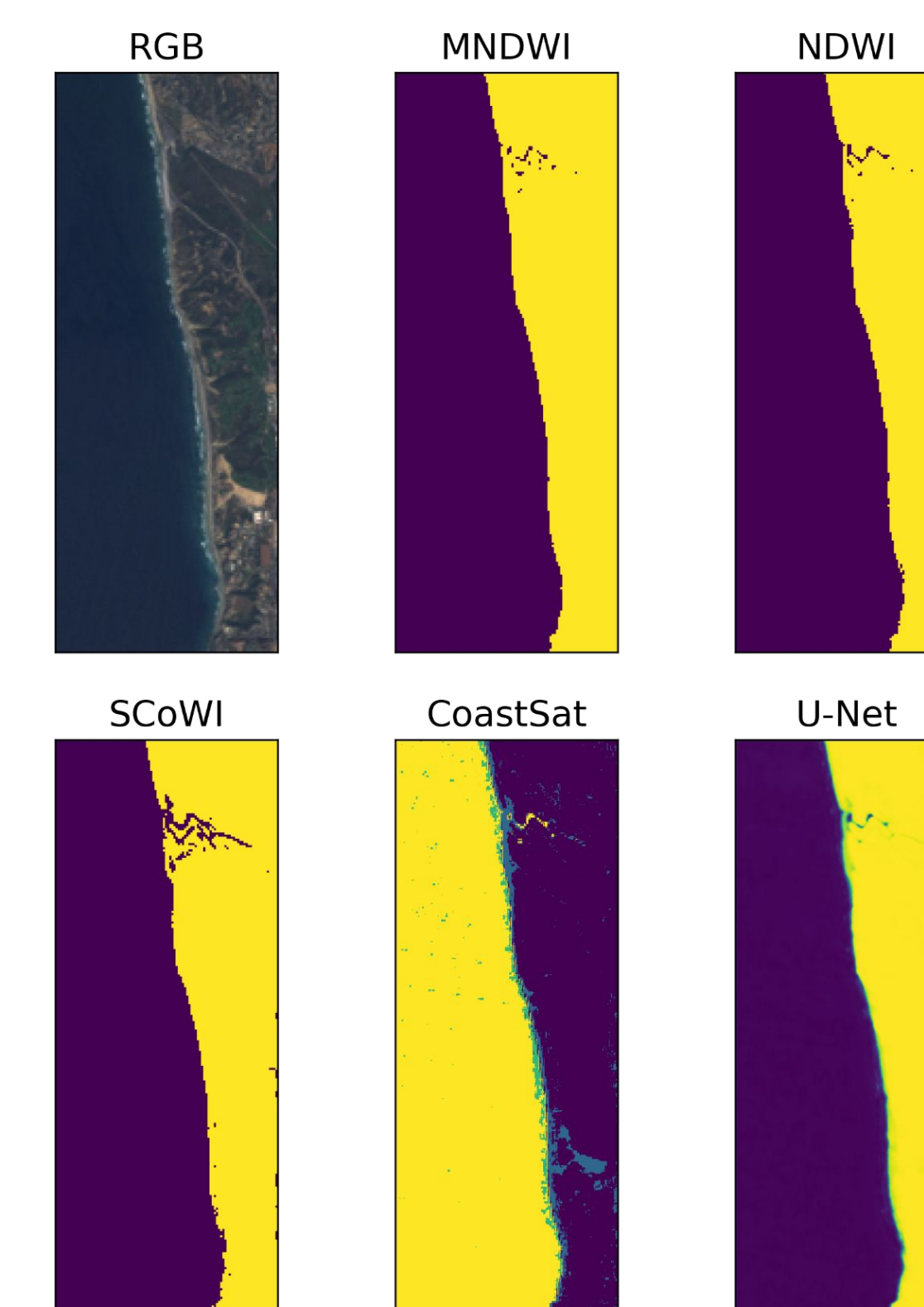


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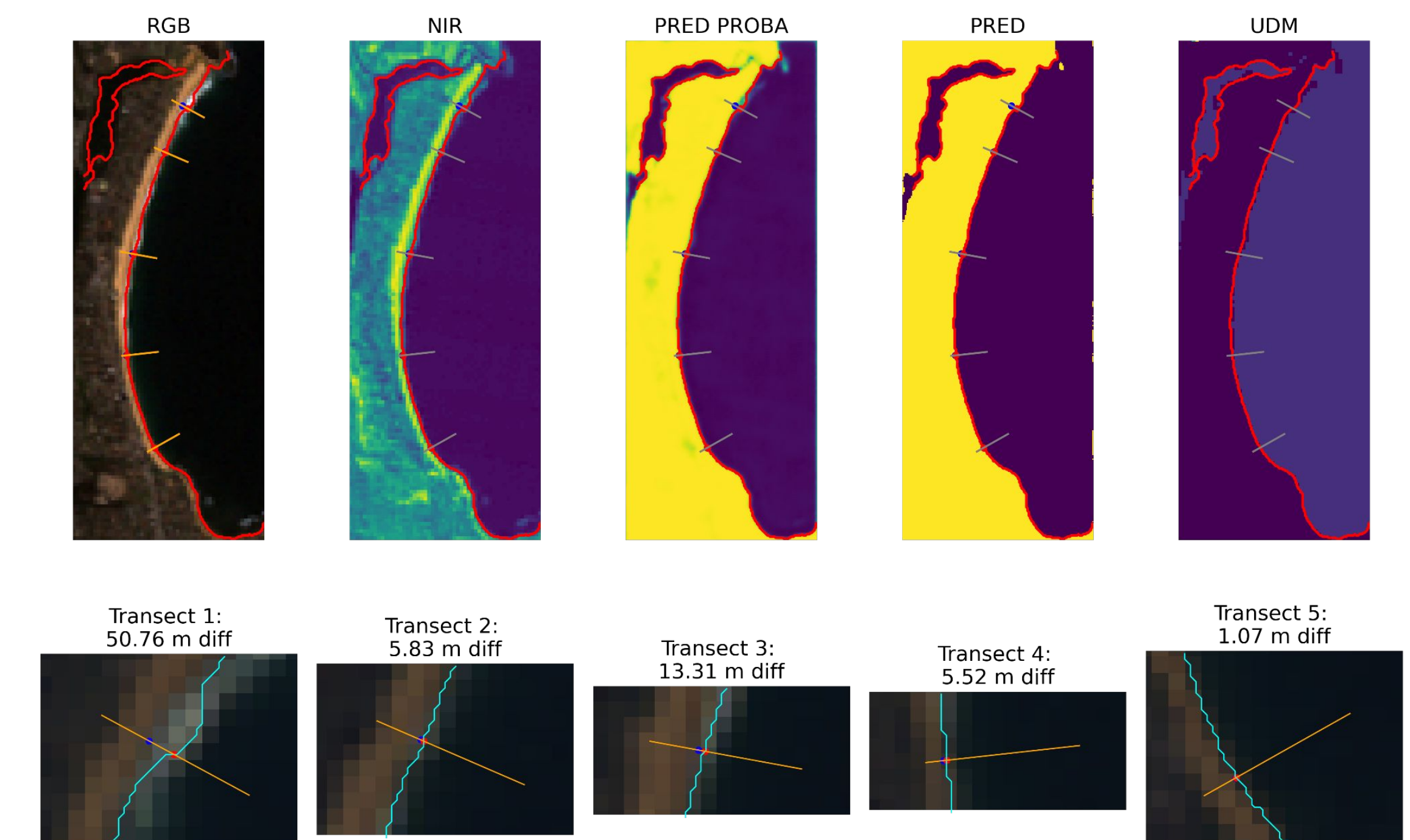


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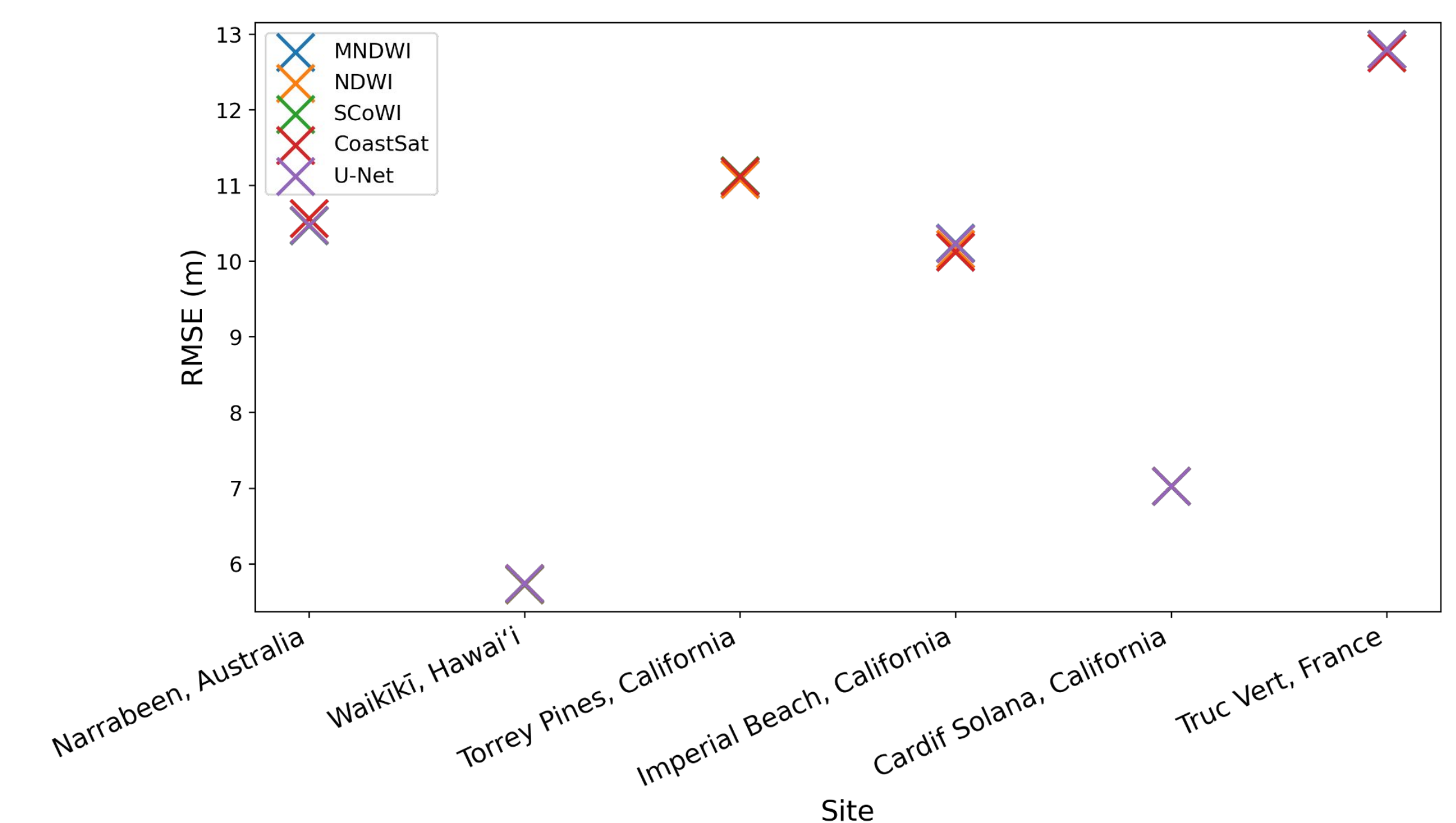


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